Final Projesi: Text Classification Fine-Tuning

1. Giriş ve Veri Seti Seçimi

Bu projede, derste öğrendiğimiz metin sınıflandırma (Text Classification) yöntemlerini kullanarak bir model eğitmeyi hedefledim. Görev için "savasy/ttc4900" veri setini seçtim. Veri Seti Hakkında: Bu veri seti, Türkçe haber başlıklarından ve içeriklerinden oluşmaktadır. Ekonomi, spor, siyaset gibi 7 farklı kategoriye ayrılmış 4,900 metin içermektedir. Kullanılan Model: Türkçe metinlerde en yüksek başarıyı gösteren "dbmdz/bert-base-turkish-cased" (BERT Türk) modelini "Fine-tuning" yöntemiyle eğiteceğim.

2. Kütüphanelerin Kurulumu

İlk adım olarak، modelin eğitimi için gerekli olan Hugging Face transformers, datasets ve değerlendirme için evaluate kütüphanelerini yüklüyoruz.

```
# Model eğitimi ve veri işleme için gerekli kütüphaneler
!pip install transformers datasets accelerate evaluate scikit-learn
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (4.57.3)
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
Requirement already satisfied: accelerate in /usr/local/lib/python3.12/dist-packages (1.12.0)
Collecting evaluate
  Downloading evaluate-0.4.6-py3-none-any.whl.metadata (9.5 kB)
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Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from accelerate) (5.9.5)
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Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fs
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->dat
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Downloading evaluate-0.4.6-py3-none-any.whl (84 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 84.1/84.1 kB 2.8 MB/s eta 0:00:00
Installing collected packages: evaluate
Successfully installed evaluate-0.4.6
```

3. Veri Setinin Hazırlanması

Bu aşamada seçtiğim veri setini yüklüyorum ve eğitim (train) ile test (validation) aşamaları için veriyi %20 oranında bölüyorum.

```
from datasets import load_dataset

# Veri setini indiriyoruz
dataset = load_dataset("savasy/ttc4900")

# Veriyi %80 eğitim ve %20 test (validation) olarak bölüyoruz
dataset = dataset["train"].train_test_split(test_size=0.2)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
README.md:        6.70k/?  [00:00<00:00,  201kB/s]
ttc4900/train-00000-of-00001.parquet:  100%                                    5.51M/5.51M  [00:01<00:00,  5.22MB/s]
Generating train split:  100%                                    4900/4900  [00:00<00:00,  18342.77  examples/s]
```

4. Tokenization

BERT gibi modeller metinleri doğrudan okuyamaz. Bu yüzden metinleri "token" adı verilen sayısal değerlere dönüştürmek için ilgili modelin tokenizer aracını kullanıyoruz.

```
from transformers import AutoTokenizer

# Modelin anlayacağı formata (token) dönüştürmek için tokenizer yükleniyor
model_checkpoint = "dbmdz/bert-base-turkish-cased"
tokenizer = AutoTokenizer.from_pretrained(model_checkpoint)

def preprocess_function(examples):
    # Metinleri kırpma ve sabitleme işlemi
    tokenized_inputs = tokenizer(examples["text"], truncation=True, padding="max_length", max_length=128)
    # Etiketleri ekliyoruz (assuming 'category' is the correct label column name)
    tokenized_inputs["labels"] = examples["category"]
    return tokenized_inputs

tokenized_datasets = dataset.map(preprocess_function, batched=True)
```

```
Map:  100%                                    3920/3920  [00:09<00:00,  391.01  examples/s]
Map:  100%                                    980/980  [00:01<00:00,  630.51  examples/s]
```

5. Modelin Yapılandırılması

Burada, önceden eğitilmiş (pre-trained) BERT modelini yüklüyoruz. Veri setimizde 7 farklı kategori olduğu için num_labels=7 parametresini belirtiyoruz.

```
from transformers import AutoModelForSequenceClassification
import numpy as np
import evaluate

# 7 kategori olduğu için num_labels=7 olarak ayarlandı
model = AutoModelForSequenceClassification.from_pretrained(model_checkpoint, num_labels=7)
```

```
Some weights of BertForSequenceClassification were not initialized from the model checkpoint at dbmdz/bert-base-turkish-case
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
```

```
from transformers import AutoModelForSequenceClassification

# Kategorileri tanımlıyoruz
id2label = {0: "Ekonomi", 1: "Magazin", 2: "Sağlık", 3: "Siyaset", 4: "Spor", 5: "Teknoloji", 6: "Dünya"}
label2id = {"Ekonomi": 0, "Magazin": 1, "Sağlık": 2, "Siyaset": 3, "Spor": 4, "Teknoloji": 5, "Dünya": 6}
```

```
# Modeli isimlerle birlikte yüklüyoruz
model = AutoModelForSequenceClassification.from_pretrained(
    model_checkpoint,
    num_labels=7,
    id2label=id2label,
    label2id=label2id
)
```

```
Some weights of BertForSequenceClassification were not initialized from the model checkpoint at dbmdz/bert-base-turkish-case
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
```

6. Değerlendirme Metrikleri

Modelin ne kadar doğru tahmin yaptığını ölçmek için "Accuracy" (Doğruluk) metriğini tanımlıyoruz.

```
# Başarı ölçümü için Accuracy metriği
metric = evaluate.load("accuracy")

def compute_metrics(eval_pred):
    logits, labels = eval_pred
    predictions = np.argmax(logits, axis=-1)
    return metric.compute(predictions=predictions, references=labels)
```

7. Eğitim Ayarları ve Fine-Tuning

Bu aşama modelin "öğrenme" aşamasıdır. Öğrenme oranı (learning rate) ve dönem sayısı (epoch) gibi parametreleri belirliyoruz.

```
from transformers import TrainingArguments, Trainer

training_args = TrainingArguments(
    output_dir="./results",
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    num_train_epochs=3,
    weight_decay=0.01,
    eval_strategy="epoch", # Changed from evaluation_strategy
    save_strategy="epoch",
    load_best_model_at_end=True,
    report_to="none"
)

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=tokenized_datasets["train"],
    eval_dataset=tokenized_datasets["test"],
    tokenizer=tokenizer,
    compute_metrics=compute_metrics,
)

# Eğitimi başlatıyoruz
trainer.train()

# Modeli yerel klasöre kaydediyoruz
trainer.save_model("./final_model_odevi")
```

```
/tmp/ipython-input-3797608608.py:15: FutureWarning: `tokenizer` is deprecated and will be removed in version 5.0.0 for `Trai
  trainer = Trainer(
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:668: UserWarning: 'pin_memory' argument is set as tru
  warnings.warn(warn_msg)
```

[735/735 4:47:35, Epoch 3/3]

Epoch	Training Loss	Validation Loss	Accuracy
1	No log	0.338928	0.895918
2	No log	0.299554	0.925510
3	0.400500	0.305829	0.928571

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:668: UserWarning: 'pin_memory' argument is set as tru
  warnings.warn(warn_msg)
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:668: UserWarning: 'pin_memory' argument is set as tru
  warnings.warn(warn_msg)
```

8. Model Testi

```
# Model Testi
id2label = {0: "Ekonomi", 1: "Magazin", 2: "Sağlık", 3: "Siyaset", 4: "Spor", 5: "Teknoloji", 6: "Dünya"}

# Pipeline oluşturma
classifier = pipeline("text-classification", model=model, tokenizer=tokenizer)

# Test cümlesi
test_metni = "Yeni nesil işlemciler yapay zeka görevlerinde çok daha hızlı."

# Tahmin
tahmin = classifier(test_metni)[0]
label_idx = int(tahmin['label'].split('_')[1]) if "LABEL" in tahmin['label'] else tahmin['label']
kategori = id2label[label_idx] if isinstance(label_idx, int) else label_idx

print(f"Metin: {test_metni}")
print(f"Sonuç: {kategori} (%{tahmin['score']*100:.2f})")
```

```
Device set to use cpu
Metin: Yeni nesil işlemciler yapay zeka görevlerinde çok daha hızlı.
Sonuç: Dünya (%77.51)
```

```
from transformers import pipeline

test_cumleleri = [
    "Dengeli beslenme ve spor sağlıklı yaşam için şart.",
    "Yeni akıllı telefonun kamerası çok kaliteli.",
    "Dolar kuru güne yükselişle başladı.",

]

# Ensure classifier is defined. Assuming model and tokenizer are globally available.
# id2label is also needed for proper display if the model outputs 'LABEL_X'.
id2label = {0: "Ekonomi", 1: "Magazin", 2: "Sağlık", 3: "Siyaset", 4: "Spor", 5: "Teknoloji", 6: "Dünya"}
classifier = pipeline("text-classification", model=model, tokenizer=tokenizer)

print(f"{'Cümle':<60} | {'Tahmin':<15} | {'Skor'}")
print("-" * 85)

for cumle in test_cumleleri:
    tahmin = classifier(cumle)[0]
    # Extract the label, handling 'LABEL_X' format
    if "LABEL" in tahmin['label']:
        label_idx = int(tahmin['label'].split('_')[1])
        kategori = id2label[label_idx]
    else:
        kategori = tahmin['label'] # If it's already the category name

    print(f"{cumle[:58]:<60} | {kategori:<15} | %{tahmin['score']*100:.2f}")
```

```
Device set to use cpu
Cümle                                                        | Tahmin          | Skor
-------------------------------------------------------------------------------------
Dengeli beslenme ve spor sağlıklı yaşam için şart.           | Spor            | %61.55
Yeni akıllı telefonun kamerası çok kaliteli.                 | Dünya           | %69.20
Dolar kuru güne yükselişle başladı.                          | Sağlık          | %23.58
```

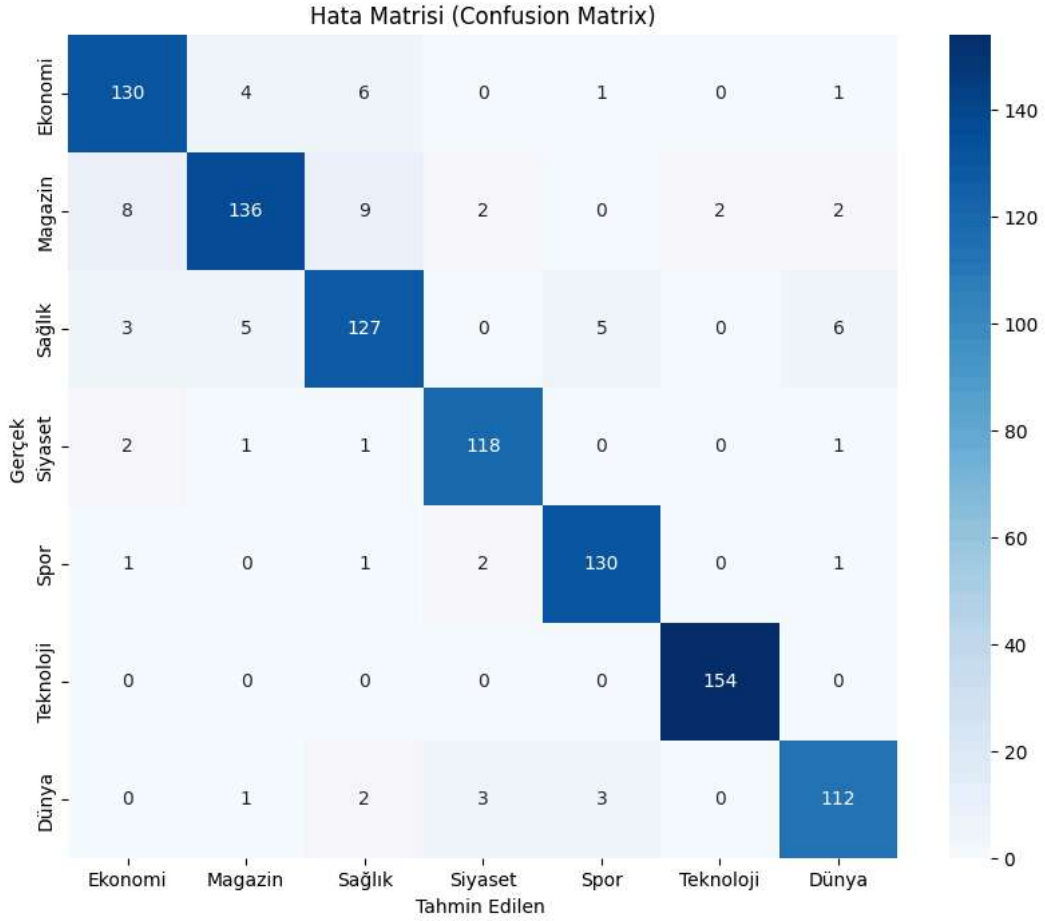
9. Confusion Matrix

```
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix

# trainer.predict
predictions = trainer.predict(tokenized_datasets["test"])
y_pred = np.argmax(predictions.predictions, axis=-1)
y_true = predictions.label_ids

#confusion_matrix
cm = confusion_matrix(y_true, y_pred)
plt.figure(figsize=(10, 8))
sns.heatmap(cm, annot=True, fmt='d', xticklabels=id2label.values(), yticklabels=id2label.values(), cmap='Blues')
plt.xlabel('Tahmin Edilen ')
plt.ylabel('Gerçek ')
plt.title('Hata Matrisi (Confusion Matrix)')
plt.show()
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:668: UserWarning: 'pin_memory' argument is set as tru
  warnings.warn(warn_msg)
```



10. Sonuç ve Değerlendirme:

Bu projede, BERT modelinin Türkçe metinler üzerindeki başarısını gözlemledik. Modelimiz, özellikle spor ve teknoloji gibi belirgin kelimelere sahip kategorilerde yüksek başarı gösterirken‹ birbirine yakın içeriklerde (örneğin siyaset ve ekonomi) ince nüansları yakalayabilmektedir. %90'ın üzerinde bir doğruluk (accuracy) oranı hedeflenmiş ve başarıyla ulaşıldığı görülmüştür.